



Monthly Running Costs

Spotly platform — infrastructure & third-party services

Prepared July 2026
Markets: Zimbabwe (+263) · Poland (+48)
Figures in USD

Planning estimates, not quotes. Prices are USD as of early 2026 and change frequently. The dominant costs are **volume-driven** (messaging, maps, compute) — the actual bill tracks user and order volume, not headcount. Payment processing is a **pass-through % of transaction value** and is listed separately, outside the monthly totals.

At a glance

Stage	Monthly users	Orders / mo	Est. monthly cost
Pilot — MVP / soft launch	~500	~2,000	\$110 – \$160
Growth	~5,000	~20,000	\$850 – \$1,100
Scale	~50,000	~150,000	\$4,800 – \$5,800

Plus fixed annual: **Apple Developer** \$99/yr (~\$8.25/mo) and **Google Play** \$25 one-time.

What Spotly runs on

Component	Technology	Billing driver
Backend API + MQTT bridge	Node.js / Express + Aedes/Mosquitto	Compute hours
Database	SQLite now → Postgres at scale	Storage + IOPS
Real-time messaging	MQTT (apps) + Socket.io (dashboard)	Compute (bundled)
OTP auth	Twilio Verify or self-managed OTP	Per verification / message
WhatsApp support	Twilio WhatsApp Business API	Per conversation / message
Push notifications	Expo Push	Free
Maps & live tracking	Google Maps / Mapbox	Map loads + directions
Image hosting	Cloudflare R2 / S3 + CDN	Storage + egress
Marketing website	Vercel / Netlify	Mostly free tier
Payments	EcoCash / card processor	% of GMV (pass-through)

Detailed breakdown by stage

PILOT ~500 users · ~2k orders/mo

Line item	Provider (example)	Cost/mo
Backend compute (1 small instance)	DigitalOcean 2GB / Railway	\$12 – \$25
Database	SQLite on the same box	\$0
MQTT broker	Mosquitto co-located	\$0
Image storage + CDN	Cloudflare R2 (few GB)	\$5
Twilio OTP (self-managed, ~1k sends)	WhatsApp/SMS send only	\$30 – \$50
WhatsApp support (user-initiated)	Meta free + Twilio msg fee	\$5 – \$15
Maps	Google/Mapbox free tier	\$0
Push notifications	Expo	\$0
Website + domain	Vercel Hobby	\$1 – \$2
Apple Developer (amortized)	\$99/yr	\$8
Subtotal		~\$110 – \$160

GROWTH ~5k users · ~20k orders/mo

Line item	Cost/mo
Backend compute (2–4 vCPU + load balancer)	\$60 – \$120
Managed Postgres	\$50 – \$100
MQTT broker (dedicated / HiveMQ Cloud)	\$25 – \$60
Image storage + CDN	\$20 – \$50
Twilio OTP (~10k verifications)	\$300 – \$500
WhatsApp support conversations	\$60 – \$120
SMS fallback	\$30 – \$60
Maps / live tracking	\$100 – \$200
Website (Vercel Pro) + monitoring (Sentry Team)	\$46
Apple Developer (amortized)	\$8
Subtotal	~\$850 – \$1,100

SCALE ~50k users · ~150k orders/mo

Line item	Cost/mo
Backend compute (multi-instance, autoscale)	\$350 – \$500
Managed Postgres (HA + read replica)	\$200 – \$400
MQTT broker (hosted cluster — EMQX/HiveMQ)	\$200 – \$500
Image storage + CDN	\$100 – \$300
Twilio OTP (~60k verifications)	\$2,000 – \$2,800
WhatsApp support conversations	\$300 – \$500
SMS fallback	\$150 – \$250
Maps / live tracking	\$400 – \$700
Website + edge + monitoring/logging	\$100 – \$190
Apple Developer (amortized)	\$8
Subtotal	~\$4,800 – \$5,800

The two costs that dominate at scale

1 · OTP verification (Twilio)

Twilio **Verify** charges ~\$0.05 per verification on top of the message delivery cost — at 60k verifications/mo that is ~\$3,000 alone.

Mitigation — already supported in the codebase. `backend/bridge/auth.js` implements a **self-managed OTP** flow that pays only the per-message send (WhatsApp authentication template ≈ \$0.01–0.03 in ZW/PL, or SMS), skipping the Verify premium. Combined with long refresh-token sessions (30 days, already configured) and preferring WhatsApp over SMS, this can cut the line by **50–70%** at scale.

2 · Maps & live tracking

Uber/Bolt-style tracking makes frequent map-load and directions calls. **Google Maps** bills per request after the free credit; **Mapbox** gives 50k free loads/mo and is often cheaper for this pattern. Mitigation: throttle driver-location redraws, cache routes, and prefer Mapbox vector tiles.

Not monthly infrastructure

Item	Nature	Notes
Payment processing	% of GMV	e.g. card ~2.9% + \$0.30; EcoCash merchant fees vary. Scales with sales.
Apple Developer Program	\$99 / year	Required to ship the iOS apps.
Google Play Console	\$25 one-time	Per developer account.
Domain registration	~\$12–20 / year	—

How to keep the bill down

1. **Start co-located** — one droplet runs Node + Mosquitto + SQLite fine for the pilot (\$12–25/mo total infra). Split services out only when load demands it.
2. **Use self-managed OTP** (`auth.js`) instead of Twilio Verify once volume grows.
3. **Prefer WhatsApp** for OTP and support — user-initiated support conversations are free under Meta's current pricing; only pay Twilio's small per-message fee.
4. **Cache aggressively** — menus/listings via MQTT retained messages (already done) and Postgres read replicas cut DB load.
5. **Move SQLite → Postgres only when needed** — the schema in `db.js` is already Postgres-compatible, so migration is a config swap, not a rewrite.

Figures are directional estimates for budgeting. Confirm live pricing with each provider (Twilio, your host, Google/Mapbox, and your payment processor) for the Zimbabwe (+263) and Poland (+48) markets before committing.